CAREER GLITCH · SHORT · SIMPLE · NO BORING PARTS

DATA SCIENTIST / ML ENGINEER IN THE AI ERA

The people who build the AI are still hired by humans — slowly, and at a high bar. This is the fastest-growing door in the building, and it still demands the degree.

★ TL;DR — Split title, split pay — print both: Data Scientist median $120,230; ML Engineer prices off the software-developer proxy at ~$136K (BLS, May 2025). The old single $135K was an ML figure misfiled over the data-scientist median. Growth for data scientists is +35% — the fastest in this entire library — with ~24,800 openings a year. The catch: enter through the data-scientist side. ML Engineer is a step-up seat, and the degree is not negotiable. ★

THE CAREER GLITCH TAKE

OUR TAKE: YES

MEDIAN ~$120K · TRAINING 2-4 years · AI RISK MEDIUM · STATE LICENSE

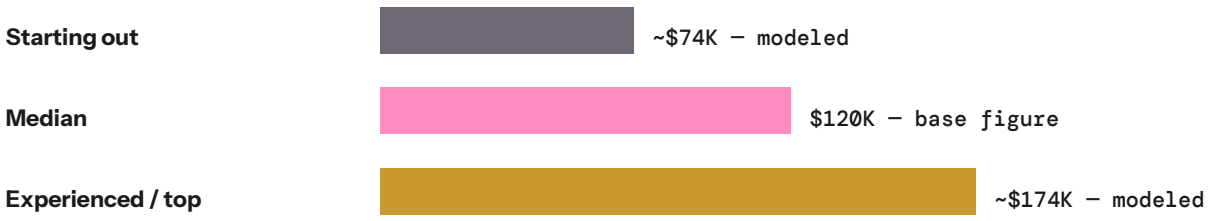
- The demand is the strongest measured anywhere in this library: +35% growth and ~24,800 openings a year for data scientists, a formal BLS entry of bachelor's degree with no experience required — and WEF ranks Big Data Specialists #1 and AI/ML Specialists #3 among fastest-growing roles. BLS names AI itself as the driver: data scientists 'will be needed to help businesses leverage AI.'

- The pay split is mandatory: $120,230 is the data-scientist median; ~$136K belongs to the ML half (software-developer proxy $135,980; research scientists $140,300). A beginner's first offers sit toward the occupation's bottom, under the $67,240 tenth-percentile line — the median is a mid-career figure on both halves.

- The junior task stack — cleaning, exploratory analysis, baseline models, routine tuning — is precisely the AutoML/LLM layer now. What gets a beginner hired is the remainder: problem framing, validation discipline and domain depth. And the master's is the competitive norm for ML and research seats: in one 2026 postings scan, graduate credentials specified in roughly two-thirds of data-scientist ads.

The site's take, built from this profession's numbers in the Career Glitch base.

THE MONEY — LOOK AT THE BARS



The Career Glitch base figure for Data Scientist / ML Engineer is a ~$120K median (compiled from U.S. Bureau of Labor Statistics, Occupational Outlook Handbook ranges). The starting and top bars are modeled around that median — orientation, not a promise. Pay swings hard by state, employer and experience: check live postings in your state before planning around any number.

WHAT DOES A DATA SCIENTIST / ML ENGINEER ACTUALLY DO?

- ✓ Frame business problems as measurable prediction or decision problems
- ✓ Clean data, build features and train models
- ✓ Evaluate models honestly — including when the answer is 'it doesn't work'
- ✓ Deploy, monitor and explain models to engineers and executives

Think: a scientist embedded in a software team — half experiments, half production code, all accountability.

WHAT AI ALREADY TOOK (AND WHAT IT CAN'T)

THE ROBOT	THE HUMAN
AutoML and AI assistants generate baseline models, boilerplate pipelines and hyperparameter searches. The 'first model in a notebook' task is fully automated — Microsoft Research scores Data Scientists 0.357 on AI applicability, #19 of the 40 most AI-overlapped occupations.	Deciding what to predict, whether the data can support it, and whether a model's errors are acceptable in the real world. Accountability for a wrong model stays human.

⚠ **Two numbers, two doors — don't let anyone merge them for you. The bachelor's is the floor (4 years; that's what the study band prices), but the ML-engineer and research seats run on a master's: budget 5–6 years total if the ML half is your target. Entry-level data-scientist postings were the LEAST common experience category in a 2026 market scan — the door is real, federal-scale, and narrower than the headline growth suggests.**

THE VIBE — IS THIS YOUR LIFE?

THE JOB IS	THE JOB IS NOT

✓ Top-of-market pay and remote work ✓ Genuinely hard, interesting problems ✓ Your skills compound — every project makes the next one faster	✗ Years of runway before the first real title ✗ Most of the work is data cleaning, not model magic ✗ You will explain statistics to executives. Forever.

Good fit? You genuinely enjoy math and code, and you can tolerate a long qualification runway for the fastest-growing ceiling in tech.
Bad fit? You want a fast pivot into tech money — this is the slowest door in the building — or a bootcamp shortcut around the degree.

THE FORK — PICK ONE DOOR

DOOR A — THE FORMAL ROUTE (DATA-SCIENTIST SIDE)	DOOR B — FROM ANALYST OR ENGINEER
A quantitative degree — computer science, statistics, math, engineering — is the door most employers screen for, and it opens the data-scientist seat first: bachelor's minimum, no experience formally required. Treat the master's as the planned second stage for ML and research seats, not a surprise discovered in year three.	Come in sideways: data analysts who learn ML properly, or software engineers who add statistics. You skip the 'junior data scientist' pile by arriving with one side already proven — and the engineer side is also the honest road toward the ML-engineer seat, which is a step-up from production software work, not a first job.

Smart move: Build end-to-end projects that are deployed and monitored, not Kaggle screenshots. One shipped system — with honest error analysis — beats ten notebooks in every interview, because validation is the skill the tools haven't taken.
Where you land after: Senior/staff data scientist or ML engineer → ML leadership, research engineering or AI product roles — entered from the data-scientist side, at the printed split: 120 to start the climb, 136+ on the ML side.

YOUR ROADMAP — 6 STEPS

1	2	3	4	5	6
AUDIT THE	ENROLL AND	PYTHON AND	CORE ML	SHIP AND	INTERVIEW AT

STEP 1 — Audit the math gap

FREE · WEEK 1

- List what the track demands: linear algebra, probability, statistics, Python.
- Choose the formal program or degree path that fills your gaps — this decision sets the whole timeline: 4 years to the floor, 5–6 to the ML-competitive tier.

STEP 2 — Enroll and start the degree track

VARIES · MONTHS 1-3

- Enter the formal program; this career's gate is educational, so get inside the pipeline early.
- Decide now whether the master's is in the plan — for ML/research seats it is the competitive norm, and sequencing it beats discovering it later.

STEP 3 — Python and data fundamentals

FREE-$200 · MONTHS 2-6

- Python, SQL, pandas and data visualization to a working standard.
- Statistics in parallel: inference, regression, experimental design — the framing skills AI doesn't do for you.

STEP 4 — Core ML properly

MONTHS 6-18

- Supervised and unsupervised learning, evaluation, leakage, class imbalance — the unglamorous core.
- Rebuild classic models from scratch once; use libraries — and AutoML — after you understand what they're doing.

STEP 5 — Ship and deploy projects

MONTHS 12-30

- Two end-to-end systems: API, monitoring, documentation, honest error analysis.
- Train validation like it's the job — because at the entry layer, it now is.

STEP 6 — Interview at the real bar, DS-side first

YEARS 2-4

- Prepare for statistics questions, ML system design and coding rounds.
- Target data-scientist roles first; treat ML-engineer postings as the step-up they are, reached with production experience.

FIRST 90 DAYS — JUST DO THIS

- □ Today: write down the formal-education gap honestly and pick the program type that closes it — bachelor's floor, master's if the ML half is the target.
- □ Days 1–30: application/enrollment steps done; Python and statistics study started daily.
- □ Days 31–60: first dataset project in progress, built around validation and error analysis, not just a model score.
- □ Days 61–90: inside the pipeline — enrolled, studying, one small end-to-end project shipped.
- □ Day 90: not a data scientist yet — but formally in the pipeline of the fastest-growing occupation in the library, with proof of work started.

NEXT — MAKE IT PERSONAL

This roadmap is the same for everyone. Your life is not.

- Not sure this career fits YOU? Take the Big Career Test — 50 questions, your top-3 matches and one honest "probably NOT you". → careerglitch.com/#test
- Already sure? Turn AI into Your Personal Career Mentor — a 2,300-word system that interviews you first, checks your hours, budget and deadline honestly, and builds YOUR plan into a Notion dashboard. → careerglitch.com/#prompt

★ MODELS ARE CHEAP. JUDGMENT IS THE DEGREE. ★

CAREER GLITCH · Figures: Career Glitch profession base (medians and training ranges compiled from U.S. Bureau of Labor Statistics, Occupational Outlook Handbook ranges). Modeled bars are estimates around the base median. License and certification rules vary by state — verify with your state board before paying for any program.